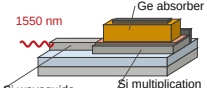
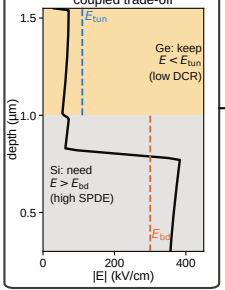
Core problem: $\max_{\mathbf{x}\in\mathcal{X}} \min_{\boldsymbol{\delta}\in\Delta} \rho(\mathbf{x},\boldsymbol{\delta})$ subject to a TCAD budget $\sum_{\ell} n_\ell c_\ell \le B$

designer ③ maximises over designs x · challenger ④ minimises over uncertainties δ · all claims checked by TCAD ①

Domain challenge

Target (1550 nm, 230 K):
SPDE ≥ η^*, DCR ≤ D^*
Design **x**: geometry, doping, V
Uncertain $\boldsymbol{\delta}$: $\boldsymbol{\theta}$, ΔV
Data: measured HWU SPADs



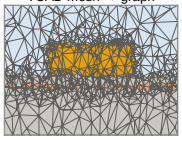


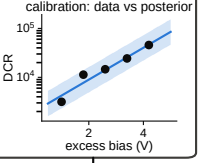
template, data

① Physics & data engine · WP1 (Xin)

Automated Silvaco TCAD: 2D screening, 3D verification. The ground truth for every claim.

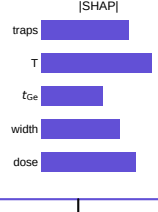
Bayesian calibration to measured devices → posterior $p(\boldsymbol{\theta}\,|\,\text{data})$ on impact-ionisation & defect parameters.





⑤ Explain · WP4

Field attribution, SHAP, symbolic regression → rules; blind 3D validation



Δ from $p(\boldsymbol{\theta}|\text{data})$

simulate $(\mathbf{x}^*, \boldsymbol{\delta}^*)$

query $(\mathbf{x}, \ell)$
2D / 3D

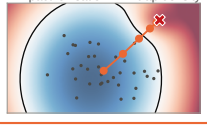
mesh graphs + fields
2D pre-train → 3D fine-tune

robust design loop

specs η^*, D^*

④ Challenger · WP3

Adversarial search over Δ
$\boldsymbol{\delta}^* = \arg\min \rho(\mathbf{x}^*, \boldsymbol{\delta})$;
estimates P_{fail};
then 250–295 K sweep

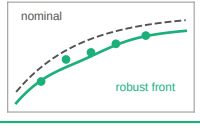


$\mathbf{x}^*$

$\boldsymbol{\delta}^*, \rho$

③ Designer · WP2

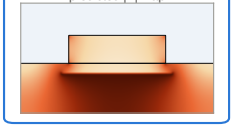
Robust multi-objective BO:
chooses next $(\mathbf{x}, \ell)$;
maximises worst-case ρ



$\mu \pm \sigma$

② Graph surrogate · WP2

GNN ensemble on mesh → field maps → physics layer → SPDE, DCR, g $(\mu \pm \sigma)$



robust designs

Outputs

- Robust front & shortlist
- Design rules
- Open toolkit
- GeSPAD-Bench
- Fab. roadmap